Digital Communication Visualization Console

Project Proposal by Team Synex

Domain: Education, Telecommunications

Problem Statement

Digital communication is a core subject in electrical and telecommunication engineering programs in Sri Lanka. However, most universities rely heavily on simulations (MATLAB/Python) to teach concepts such as source coding, packetization, CRC, and channel errors.

Students often understand mathematical theory but struggle to visualize:

- How variable-length coding compresses data
- How packets are formed
- How CRC bits are generated
- How noise corrupts bits
- How corrupted packets are detected and discarded

Therefore, there is a need for an affordable, locally developed, hardware-based communication system demonstrator that visually represents each stage of a digital communication chain.

Stakeholder Analysis

Primary Stakeholders

- State universities offering Telecommunication degrees
- Private engineering institutions
- Technical and vocational training institutes
- Institutions such as University of Moratuwa and University of Peradeniya conduct communication systems laboratories where this device can be directly deployed

Secondary Stakeholders

- Lecturers who require demonstrative teaching tools
- Undergraduate students learning digital communications
- Lab coordinators managing equipment budgets

Stakeholder Insights

Stakeholders Consulted - Dr.Samiru Gayan, Senior Lecturer, Electronic and Telecommunications Department, University of Moratuwa.

During preliminary discussions with Dr. Samiru Gayan, it was highlighted that current laboratory sessions rely heavily on simulations. Although simulations are useful, they do not provide a clear physical representation of:

- Bitstream formation
- Packet boundaries
- Redundancy addition
- Bit corruption due to noise
- Packet rejection mechanisms

Proposed Solution

The Digital Communication Visualization Console (DCVC) is a desktop unit that demonstrates:

- Symbol generation
- Fixed-length and Huffman source coding
- Packetization (4–8 bit packets)
- Parity and CRC error detection
- Adjustable noisy channel
- Packet validation and rejection

Users can enter short messages, observe LED-displayed codewords, divide the bitstream into small packets (4, 6, or 8 bits), add CRC/parity bits, transmit through a simulated noisy channel, and observe accepted or rejected packets.

Technical Specifications

Parameter	**Specification**
Symbols	5 (A–E)
Coding Methods	Fixed (3-bit), Huffman
Packet Sizes	4, 6, 8 bits
Error Detection	Even/Odd Parity, CRC
CRC Divisor	3–4 bit user input
Noise Model	Adjustable BER
Controller	ESP32
Display	2.4” TFT LCD
Bit Visualization	8–16 LED array
Power Supply	5V DC

Table 1: DCVC Technical Specifications

Budget Estimation

Category	Item	Total (LKR)
Microcontroller	ESP32	1,000
Display	2.4" TFT LCD	1,700
LED Arrays & Drivers	74HC595 + LEDs	3,500
Input Controls	Push Buttons, Encoder, DIP Switches	2,500
Power & Regulation	5V Supply, Regulators	2,500
PCB Fabrication	2-layer PCB	5,000
Enclosure	Acrylic/ABS casing	5,000
Miscellaneous	Wiring, Connectors, Mounting	3,000
Estimated Total		**24,200**

Table 2: DCVC Budget Breakdown

According to this approximation, the total cost per unit is around LKR 25,000. The expected selling price could be LKR 30,000–35,000.

Hardware and Software Complexity

The system consists of four main subsystems:

- **Input Subsystem** – Push buttons (A–E), probability adjustment knob, packet size selector, CRC divisor input.
- **Processing Subsystem** – ESP32 microcontroller implementing: Huffman tree generation, Fixed-length coding, Packetization, CRC computation, Pseudo-random noise injection.
- **Visualization Subsystem** – LED arrays for codewords and packet bits, TFT display for packet view.
- **Control & Power Subsystem** – Voltage regulation, LED driver circuits, protection components.

Estimated Market Requirement in Sri Lanka

Estimated institutions:

- ≈10 universities with engineering programs,
- ≈20 technical institutes

Assuming:

- 4–5 units per university,
- 2–3 units per technical institute

Total potential demand ≈ 80–100 units
Estimated annual demand ≈ 15–25 units

This makes the product commercially viable within the educational sector.